

PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering
Jackfruit Mini Project - Software Engineering (UE24CS341A)

Software Requirements Specification
for Project Management System

Field	Details
Project Title	Project Management System
Group Number	Team 14
Version	1.0 approved
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Organization	PES University, Department of Computer Science and Engineering
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Revision History

Name	Date	Reason For Changes	Version
Team 14	05 September 2026	Initial version of the Software Requirements Specification	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for Project Management System, a web based application that allows a team to organize projects, manage tasks on a Kanban board, plan Agile sprints, track time and view project progress through a live analytics dashboard. This document covers the complete first version of the system and is intended to be the single reference for what the system must do.

1.2 Intended Audience and Reading Suggestions

This document is intended for the development team building the system, the course instructor evaluating the project, and any reviewer who needs to understand the expected behaviour of the system. Section 2 gives an overall picture of the product. Section 3 covers external interfaces. Section 4 covers the analysis models. Section 5 lists the detailed functional requirements grouped by feature. Section 6 covers non functional, security and business rule requirements. The appendices contain the glossary, field layout and traceability matrix.

1.3 Product Scope

Project Management System is a new, self contained web application. It allows a team to create workspaces and projects, break work into tasks and subtasks, plan work using an Agile backlog and sprints, log time against tasks, collaborate through comments and a live activity feed, and monitor progress through an analytics dashboard. Task updates are pushed to connected users in real time. The system is being built as a course mini project and is scoped for use by a small team rather than a large organisation.

1.4 References

- Jackfruit - Software Engineering Project Guidelines, PES University, 2026
- Software Engineering - Level 3 Mini Project Student Submission Guide, PES University
- Team 14 Synopsis / Project Proposal document

2. Overall Description

2.1 Product Perspective

Project Management System is a new, standalone product and is not a follow on to an existing system. It consists of a React based frontend, a Django REST Framework backend, a PostgreSQL database, and a WebSocket layer built using Django Channels for pushing live updates to connected clients. The frontend communicates with the backend using a REST API for standard operations and a WebSocket connection for real time task and activity updates.

2.2 Product Functions

- User registration, login and role based access control within a workspace
- Creation and management of workspaces and projects
- Task and subtask creation, assignment and Kanban board management
- Task dependency management with cycle detection
- Agile backlog and sprint planning with story point estimation
- Time logging against tasks
- Real time task board updates using WebSockets
- Comments, mentions and a live activity feed
- In app notifications
- Analytics dashboard showing burndown, velocity and workload

2.3 User Classes and Characteristics

User Class	Privileges	Characteristics
Owner	Full control of a workspace, including deleting it and assigning any role	Sets up the workspace, typically one per workspace
Admin	Manage projects, members and roles other than Owner	Manages day to day workspace operations
Member	Create and manage tasks assigned to them, log time, comment	Regular contributor, most frequent user of the system
Viewer	Read only access to projects and dashboards	Stakeholder who needs visibility but does not edit content

2.4 Operating Environment

The system is a web application accessed through a modern browser such as Chrome, Firefox or Edge, on desktop or laptop devices. The backend runs on Python 3 with Django REST Framework and Django Channels, deployed on a Linux based server or container. The database is PostgreSQL. The frontend is a React single page application. A Redis instance is used as the channel layer backing for WebSocket connections.

2.5 Design and Implementation Constraints

- The backend must be implemented using Django REST Framework in Python
- Real time updates must be implemented using Django Channels over WebSockets
- The frontend must be implemented using React
- PostgreSQL must be used as the primary database
- The project must follow Agile methodology with sprint based delivery
- Git and GitHub must be used for version control, with CI/CD through Jenkins
- The project must be completed within the timeline defined by the course schedule

2.6 Assumptions and Dependencies

- Users are assumed to have a modern browser and a stable internet connection
- Third party libraries such as React, Django REST Framework and Django Channels remain available and compatible during development
- The system depends on a Redis instance being available for the WebSocket channel layer
- The project assumes a small team of users for demonstration purposes rather than large scale production load

3. External Interface Requirements

3.1 User Interfaces

The system provides a responsive web interface with the following primary screens: login and registration, workspace and project dashboard, Kanban task board, backlog and sprint planning screen, task detail panel with comments and activity feed, analytics dashboard, and a notification panel. Every screen follows a consistent layout with a persistent navigation sidebar and a header showing the current workspace and project.

3.2 Software Interfaces

- Django REST API exposing JSON endpoints consumed by the React frontend
- PostgreSQL database accessed through the Django ORM
- Redis used as the channel layer for Django Channels WebSocket connections

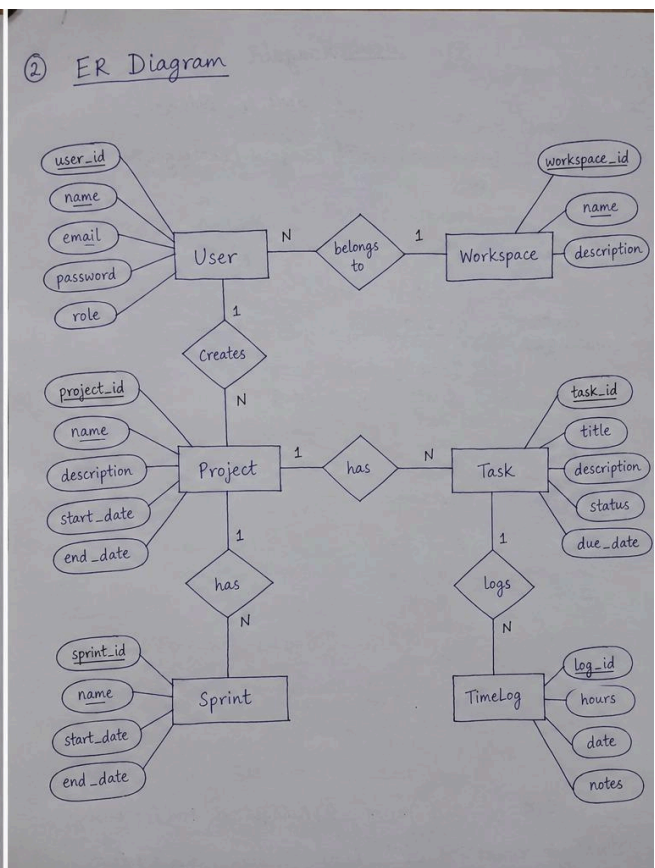
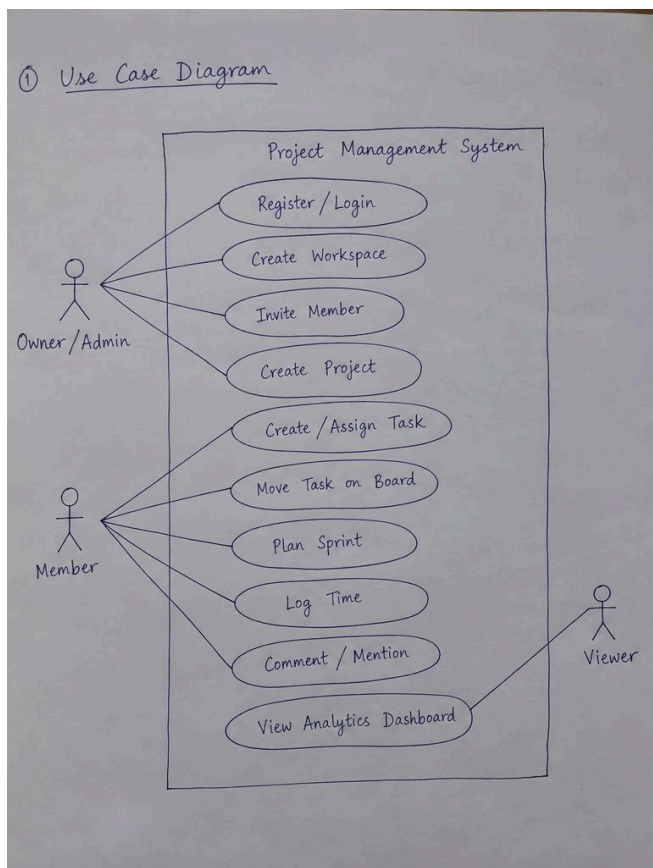
3.3 Communications Interfaces

- HTTPS is used for all standard REST API requests and responses
- WSS (WebSocket Secure) is used for real time task board and notification updates
- All API responses use JSON as the data format

4. Analysis Models

The table below lists the primary use cases supported by the system, along with the primary actor, precondition and expected outcome for each. A corresponding use case diagram and class diagram are maintained separately as part of the design documentation.

Use Case ID	Use Case	Primary Actor	Precondition	Outcome
UC-01	Register and Login	Any User	Application available	Authenticated session created
UC-02	Create Workspace	Owner	User logged in	New workspace created
UC-03	Invite Member	Owner / Admin	Workspace exists	Member added with assigned role
UC-04	Create Project	Member	Workspace exists	New project created
UC-05	Create and Assign Task	Member	Project exists	Task created and assigned
UC-06	Move Task on Board	Member	Task exists	Task status updated in real time
UC-07	Plan Sprint	Member	Backlog has items	Sprint created with assigned items
UC-08	Log Time	Member	Task assigned to user	Time entry recorded
UC-09	Comment and Mention	Member	Task exists	Comment saved, mentioned user notified
UC-10	View Analytics Dashboard	Owner / Admin / Member	Project has data	Dashboard displayed with current metrics



5. System Features

5.1 Authentication, Workspace and Access Control

5.1.1 Description and Priority

Handles user registration, login, workspace creation, member invitations and role based access control.

Priority: High. Owned by Samarth M H.

5.1.2 Stimulus/Response Sequences

- User submits registration details, system creates account and confirms success
- User submits login credentials, system validates and returns an authentication token
- Owner submits an invite request, system generates a unique invite link
- Owner or Admin changes a member role, system updates permissions immediately

5.1.3 Functional Requirements

ID	Requirement	Priority	Verification
FR-AUTH-01	The system shall allow a new user to register with a unique email address and password, and allow a registered user to log in and receive an authentication token.	High	Functional test
FR-AUTH-02	The system shall allow an authenticated user to create a new workspace or organization.	High	Functional test
FR-AUTH-03	The system shall allow a workspace Owner to invite members using a unique invite link.	High	Functional test
FR-AUTH-04	The system shall allow an Owner or Admin to assign the role Owner, Admin, Member or Viewer to a member, and shall restrict workspace and project actions based on the assigned role.	High	Authorization test

5.2 Project and Task Management

5.2.1 Description and Priority

Handles creation and management of projects, tasks, subtasks and the Kanban board. Priority: High. Owned by Sameeksha Prasanna Kumar.

5.2.2 Stimulus/Response Sequences

- Member creates a project, system adds it to the workspace
- Member creates a task with details, system adds it to the board in the default status column
- User drags a task card to another column, system updates its status
- Member defines a dependency between two tasks, system validates it before saving

5.2.3 Functional Requirements

ID	Requirement	Priority	Verification
FR-TASK-01	The system shall allow an authenticated member to create a project within a workspace.	High	Functional test
FR-TASK-02	The system shall allow a project member to create a task, break it into subtasks, and assign it to one project member.	High	Functional test
FR-TASK-03	The system shall display tasks on a Kanban board organized by status column, and allow a user to move a task between columns using drag and drop.	High	Functional test
FR-TASK-04	The system shall allow a dependency to be defined between two tasks and shall detect and reject a dependency that would create a cycle.	Medium	Negative test

5.3 Agile Sprint and Time Management

5.3.1 Description and Priority

Handles the product backlog, sprint planning, story point estimation and time logging. Priority: High. Owned by Sanjana Guttapally.

5.3.2 Stimulus/Response Sequences

- Member adds an item to the backlog, system stores it as unassigned to any sprint
- Member creates a sprint and moves backlog items into it, system associates the items with the sprint
- Member logs time against a task, system stores the entry against that task and user
- System recalculates the burndown and velocity charts whenever a task status or estimate changes

5.3.3 Functional Requirements

ID	Requirement	Priority	Verification
FR-SPR-01	The system shall allow a project member to add an item to the product backlog and assign it a story point estimate.	High	Functional test
FR-SPR-02	The system shall allow a project member to create a sprint with a defined start and end date and move backlog items into it.	High	Functional test
FR-SPR-03	The system shall allow a user to log time spent against a task assigned to them.	High	Functional test
FR-SPR-04	The system shall generate a burndown chart showing remaining story points across an active sprint and a velocity chart showing story points completed per sprint.	Medium	Functional test

5.4 Real Time Collaboration, Notifications and Analytics

5.4.1 Description and Priority

Handles live task board updates, comments, mentions, the activity feed, in app notifications and the analytics dashboard. Priority: High. Owned by Sanapureddy Hemanth Kumar Reddy.

5.4.2 Stimulus/Response Sequences

- A user updates a task, system pushes the change to all connected members of that project over WebSocket
- A user posts a comment mentioning a teammate, system saves the comment and notifies the mentioned user
- A user opens the analytics dashboard, system computes and displays current workload and progress metrics

5.4.3 Functional Requirements

ID	Requirement	Priority	Verification
FR-RTC-01	The system shall push a task update to all connected project members in real time using a WebSocket connection.	High	Integration test
FR-RTC-02	The system shall allow a user to add a comment to a task and mention another member within a comment.	High	Functional test
FR-RTC-03	The system shall record task and comment activity in a project activity feed and generate an in app notification when a user is mentioned or assigned a task.	High	Functional test
FR-RTC-04	The system shall display an analytics dashboard showing workload distribution and overdue task counts across project members.	Medium	Functional test

6. Other Nonfunctional Requirements

6.1 Performance Requirements

ID	Category	Measurable Requirement	Verification
NFR-001	Performance	95 percent of task board requests shall receive a response within 2 seconds under normal load.	Performance test
NFR-002	Availability	The application shall be available at least 99 percent during the defined operating window, excluding planned maintenance.	Operational review
NFR-003	Usability	A new user shall be able to create and assign a task in no more than 4 primary screens.	Usability walkthrough
NFR-004	Reliability	A WebSocket disconnection shall not cause loss of task data already saved to the database.	Failure/reconnect test
NFR-005	Maintainability	Business logic shall be separated into identifiable modules with automated unit tests.	Code review
NFR-006	Scalability	The system shall support at least 50 concurrent WebSocket connections per project without noticeable delay in updates.	Load test

6.2 Safety Requirements

- The system shall require explicit confirmation before permanently deleting a workspace or project
- The system shall prevent permanent loss of task data by soft deleting tasks before final removal

6.3 Security Requirements

ID	Security Requirement	Validation Approach
SEC-001	Passwords shall not be stored in plaintext.	Code and configuration review
SEC-002	Role based access shall prevent a Viewer from creating or editing tasks.	Authorization test
SEC-003	The system shall validate and sanitize user supplied input before processing.	Negative and security test
SEC-004	Authenticated API and WebSocket connections shall reject expired or invalid tokens.	Security test
SEC-005	A member removed from a workspace shall immediately lose access to that workspace's data.	Authorization test
SEC-006	Security relevant events such as failed login attempts shall be logged.	System test

6.4 Software Quality Attributes

- Maintainability: modules for auth, tasks, sprints and collaboration shall be independently testable
- Portability: the backend shall run on any Linux based environment supporting Python 3 and PostgreSQL
- Reusability: the REST API shall be documented so it can be reused by a future mobile client
- Robustness: the system shall handle invalid input without crashing, returning a clear error response instead

6.5 Business Rules

- Only a workspace Owner can delete a workspace
- A task cannot be marked Done while it has an incomplete dependency
- Only the assigned member or a project Owner can log time against a task
- A sprint cannot be started if it has no backlog items assigned to it
- Only an Owner or Admin can invite or remove a workspace member
- A Viewer has read only access and cannot create, edit or delete any content

7. Other Requirements

- The database shall retain task and activity history for the duration of the project for reporting purposes
- The system shall be deployable using Docker containers for both the backend and frontend
- The system is intended for a single institution deployment and does not require multi tenant data isolation beyond workspace level separation

8. Validation Specification (Initial Planning)

At this stage the validation specification is planning focused. It explains how each category of requirement will eventually be validated once implementation is complete. This is not the final validation or test report, which will be produced after system testing.

Requirement Category	Planned Validation	Example
Functional	Black box functional testing of each feature	Creating a task and moving it across the Kanban board
Security	Authentication and authorization testing, basic penetration testing	A Viewer attempting to access an admin only endpoint
Performance	Load and response time testing	Concurrent task board requests from multiple users
Usability	Task based walkthrough with a new user	A new user creating and assigning a task
Reliability	Failure and retry scenario testing	A WebSocket disconnecting and reconnecting mid update
Unit	Automated unit tests with coverage reporting	Story point total calculation for a sprint

9. Project Plan

The project follows an Agile lifecycle delivered across six sprints. The table below breaks the work down by package, with owner and estimated effort in person days.

Work Package	Tasks	Owner	Estimated Effort (Person Days)	Output
WP1	Requirements gathering and use case definition	All	4	SRS
WP2	Architecture and UML design	Samarth / Sameeksha	4	Architecture and design document
WP3	Auth, workspace and access control module	Samarth	7	Working feature
WP4	Project and task management module	Sameeksha	7	Working feature
WP5	Agile sprint and time management module	Sanjana	7	Working feature
WP6	Real time collaboration, notifications and analytics module	Hemanth	8	Working feature
WP7	Testing, CI/CD and security validation	All	6	Test and validation evidence
WP8	Final validation, documentation and report	All	4	Final report

Total estimated effort across the team is approximately 48 person days, distributed across the six sprint schedule described in the project synopsis.

10. Tool Stack

Activity	Tool / Technology	Evidence to Submit
Version control	Git and GitHub	Repository URL, commit history
Agile planning	GitHub Projects / Jira	Backlog, user stories, sprint board
CI/CD	Jenkins	Pipeline screenshot or log
Containerization	Docker	Dockerfile and build/run evidence
Testing	Pytest, Jest	Test results
Static analysis	SonarQube or similar	Analysis report
Real time layer	Django Channels with Redis	WebSocket connection logs
Design	Draw.io	UML diagrams

Appendix A: Glossary

Term	Definition
Workspace	A top level container that groups projects and members, owned by a single Owner
Project	A collection of tasks within a workspace, worked on by assigned members
Task	A unit of work within a project that can be assigned, tracked and moved across board columns
Subtask	A smaller unit of work that belongs to a parent task
Backlog	A list of work items that have not yet been assigned to a sprint
Sprint	A fixed time period during which a defined set of backlog items are worked on
Story Point	A relative unit used to estimate the effort required to complete a backlog item
Burndown Chart	A chart showing remaining story points in a sprint over time
Velocity	The average number of story points a team completes per sprint
RBAC	Role Based Access Control, restricting actions based on assigned user role
WebSocket	A persistent connection protocol used to push real time updates from server to client
Kanban Board	A visual board that displays tasks organized into status columns

Appendix B: Field Layouts

The table below lists the primary fields used across the core entities of the system.

Field	Length	Data Type	Description	Mandatory
user_id	36	UUID	Unique identifier for a user	Y
email	100	String	Unique login email address	Y
password_hash	128	String	Hashed user password	Y
workspace_id	36	UUID	Unique identifier for a workspace	Y
role	20	Enum	Owner, Admin, Member or Viewer	Y
project_id	36	UUID	Unique identifier for a project	Y
task_id	36	UUID	Unique identifier for a task	Y
task_title	150	String	Title of the task	Y
status	20	Enum	Backlog, To Do, In Progress, Done	Y

Field	Length	Data Type	Description	Mandatory
story_points	5	Integer	Estimated effort for a backlog item	N
sprint_id	36	UUID	Unique identifier for a sprint	N
time_logged_minutes	10	Integer	Time logged against a task in minutes	N
comment_text	1000	String	Content of a comment on a task	Y

Appendix C: Requirement Traceability Matrix

At this stage of the project, architecture, design and code references are marked as TBD and will be completed once the corresponding design and implementation artifacts are produced.

Req ID	Architecture Ref	Design Ref	Code/File Ref	Test Case ID	System Test ID
FR-AUTH-01	TBD	TBD	TBD	TBD	TBD
FR-AUTH-02	TBD	TBD	TBD	TBD	TBD
FR-AUTH-03	TBD	TBD	TBD	TBD	TBD
FR-AUTH-04	TBD	TBD	TBD	TBD	TBD
FR-TASK-01	TBD	TBD	TBD	TBD	TBD
FR-TASK-02	TBD	TBD	TBD	TBD	TBD
FR-TASK-03	TBD	TBD	TBD	TBD	TBD
FR-TASK-04	TBD	TBD	TBD	TBD	TBD
FR-SPR-01	TBD	TBD	TBD	TBD	TBD
FR-SPR-02	TBD	TBD	TBD	TBD	TBD
FR-SPR-03	TBD	TBD	TBD	TBD	TBD
FR-SPR-04	TBD	TBD	TBD	TBD	TBD
FR-RTC-01	TBD	TBD	TBD	TBD	TBD
FR-RTC-02	TBD	TBD	TBD	TBD	TBD
FR-RTC-03	TBD	TBD	TBD	TBD	TBD
FR-RTC-04	TBD	TBD	TBD	TBD	TBD
SEC-001	TBD	TBD	TBD	TBD	TBD
SEC-002	TBD	TBD	TBD	TBD	TBD
SEC-003	TBD	TBD	TBD	TBD	TBD
SEC-004	TBD	TBD	TBD	TBD	TBD
SEC-005	TBD	TBD	TBD	TBD	TBD
SEC-006	TBD	TBD	TBD	TBD	TBD